Mobile Communications
Chapter 3 : Media Access

- Motivation
- SDMA, FDMA, TDMA
- Aloha
- Reservation schemes
- Collision avoidance, MACA
- Polling
- CDMA
- SAMA
- Comparison

Motivation

Can we apply media access methods from fixed networks?

Example CSMA/CD

- **C**arrier **S**ense **M**ultiple **A**ccess with **C**ollision **D**etection
- send as soon as the medium is free, listen into the medium if a collision occurs (original method in IEEE 802.3)

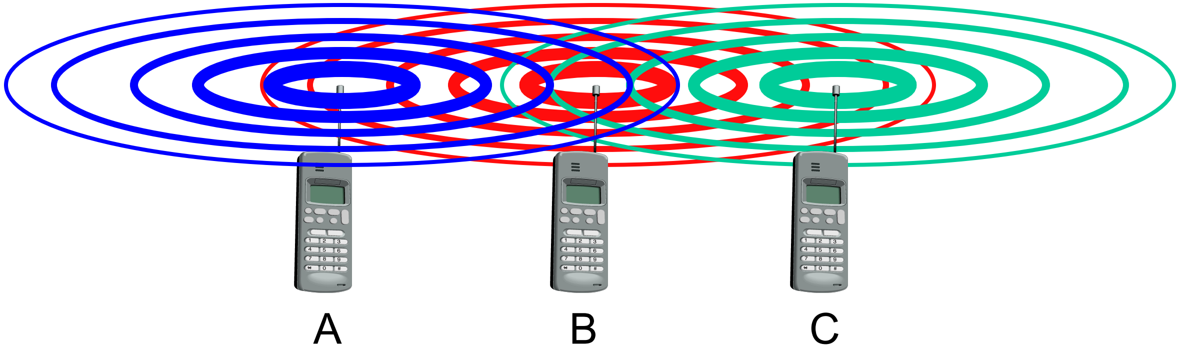
Problems in wireless networks

- signal strength decreases proportional to the square of the distance
- the sender would apply CS and CD, but the collisions happen at the receiver
- it might be the case that a sender cannot “hear” the collision, i.e., CD does not work
- furthermore, CS might not work if, e.g., a terminal is “hidden”

Motivation - hidden and exposed terminals

Hidden terminals

- A sends to B, C cannot receive A
- C wants to send to B, C senses a “free” medium (CS fails)
- collision at B, A cannot receive the collision (CD fails)
- A is “hidden” for C



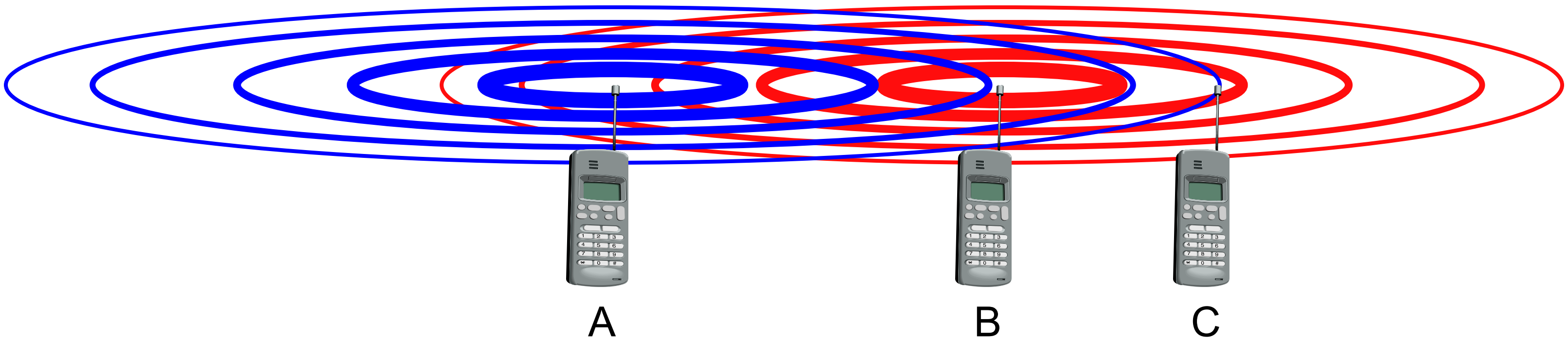
Exposed terminals

- B sends to A, C wants to send to another terminal (not A or B)
- C has to wait, CS signals a medium in use
- but A is outside the radio range of C, therefore waiting is not necessary
- C is “exposed” to B

Motivation - near and far terminals

Terminals A and B send, C receives

- signal strength decreases proportional to the square of the distance
- the signal of terminal B therefore drowns out A's signal
- C cannot receive A



If C for example was an arbiter for sending rights, terminal B would drown out terminal A already on the physical layer

Also severe problem for CDMA-networks - precise power control needed!

Access methods SDMA/FDMA/TDMA

SDMA (Space Division Multiple Access)

- segment space into sectors, use directed antennas
- cell structure

FDMA (Frequency Division Multiple Access)

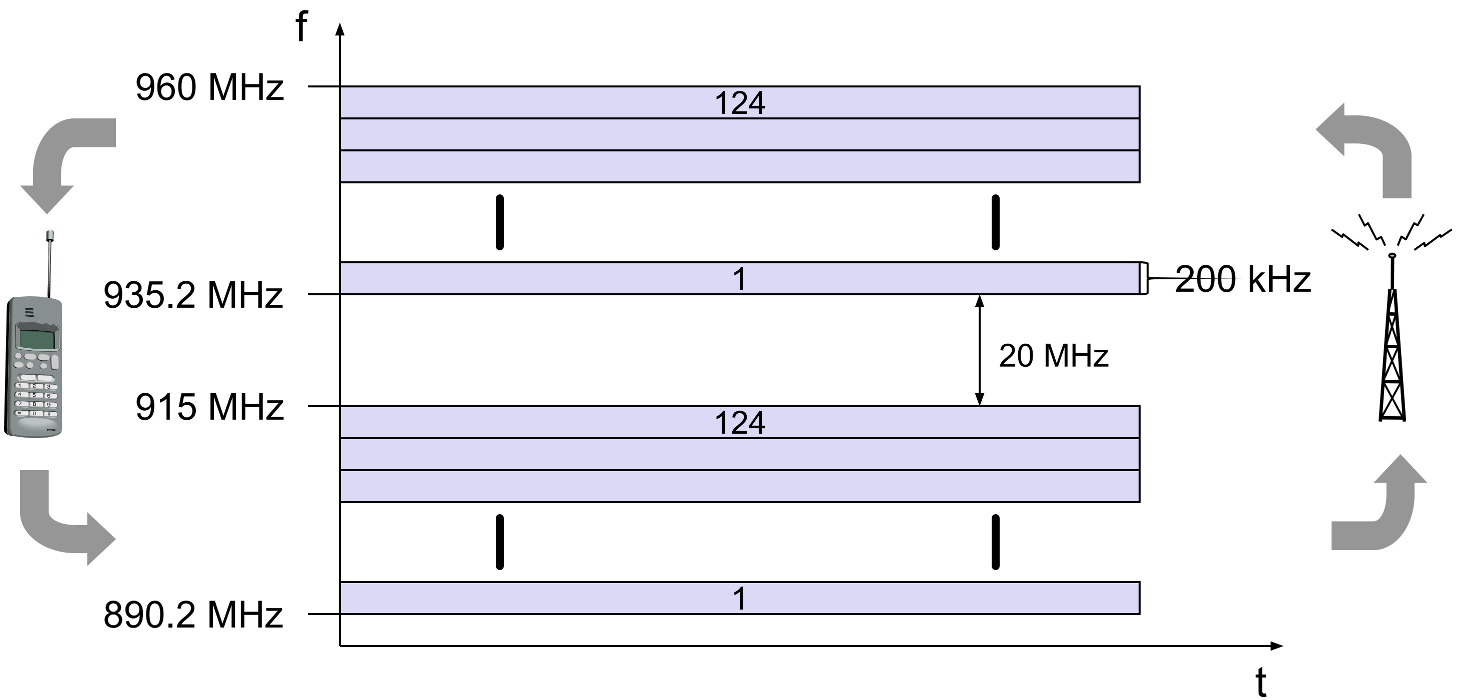
- assign a certain frequency to a transmission channel between a sender and a receiver
- permanent (e.g., radio broadcast), slow hopping (e.g., GSM), fast hopping (FHSS, Frequency Hopping Spread Spectrum)

TDMA (Time Division Multiple Access)

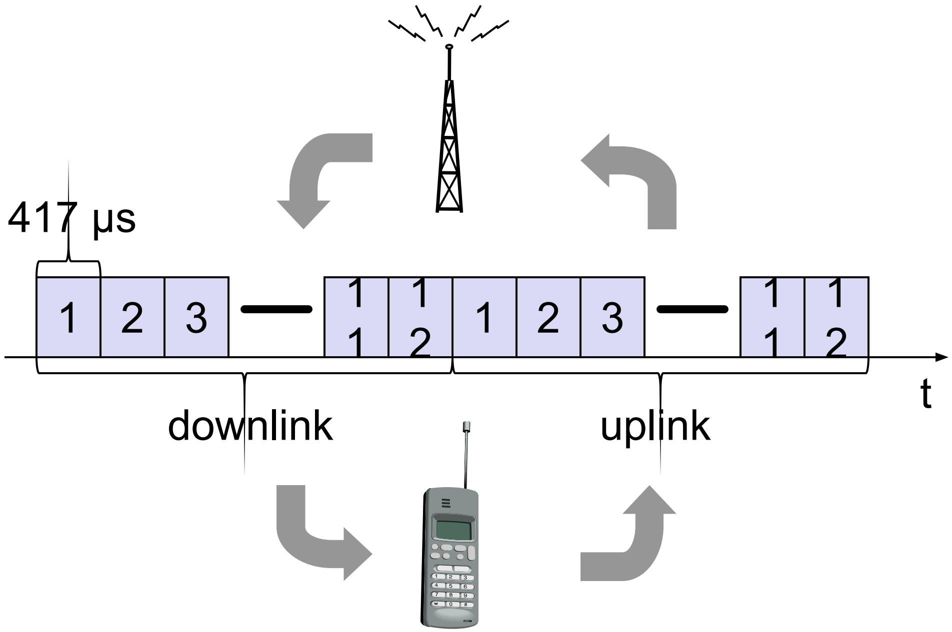
- assign the fixed sending frequency to a transmission channel between a sender and a receiver for a certain amount of time

The multiplexing schemes presented in chapter 2 are now used to control medium access!

FDD/FDMA - general scheme, example GSM



TDD/TDMA - general scheme, example DECT

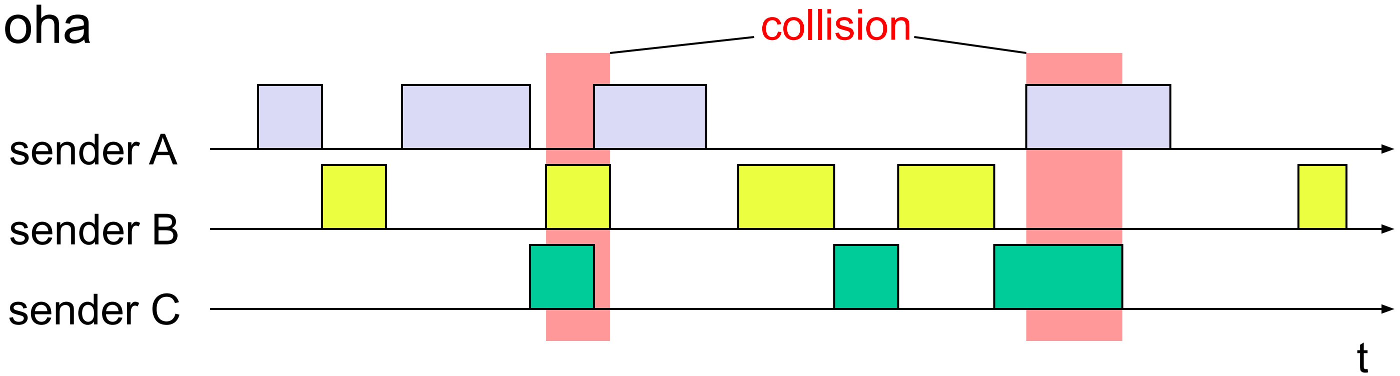


Aloha/slotted aloha

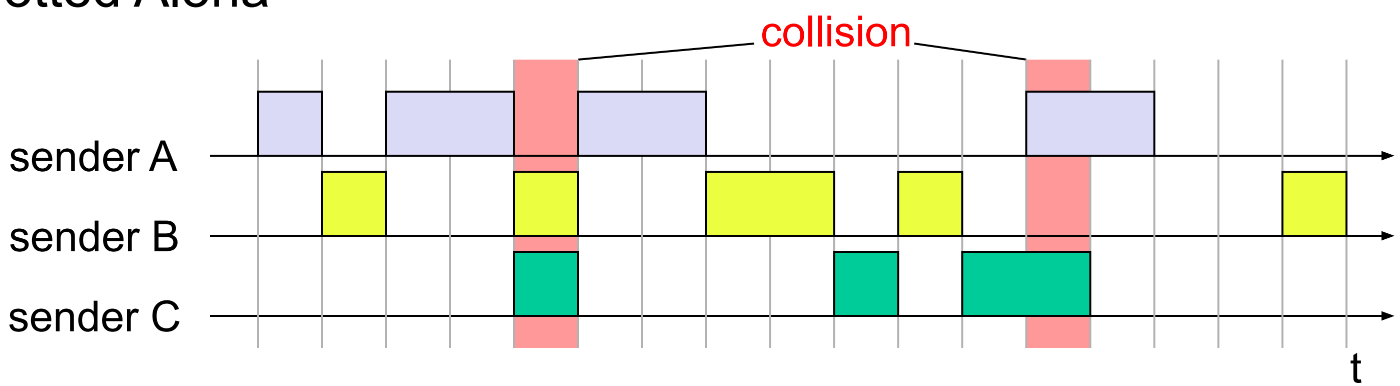
Mechanism

- random, distributed (no central arbiter), time-multiplex
- Slotted Aloha additionally uses time-slots, sending must always start at slot boundaries

Aloha



Slotted Aloha



DAMA - Demand Assigned Multiple Access

Channel efficiency only 18% for Aloha, 36% for Slotted Aloha (assuming Poisson distribution for packet arrival and packet length)

Reservation can increase efficiency to 80%

- a sender *reserves* a future time-slot
- sending within this reserved time-slot is possible without collision
- reservation also causes higher delays
- typical scheme for satellite links

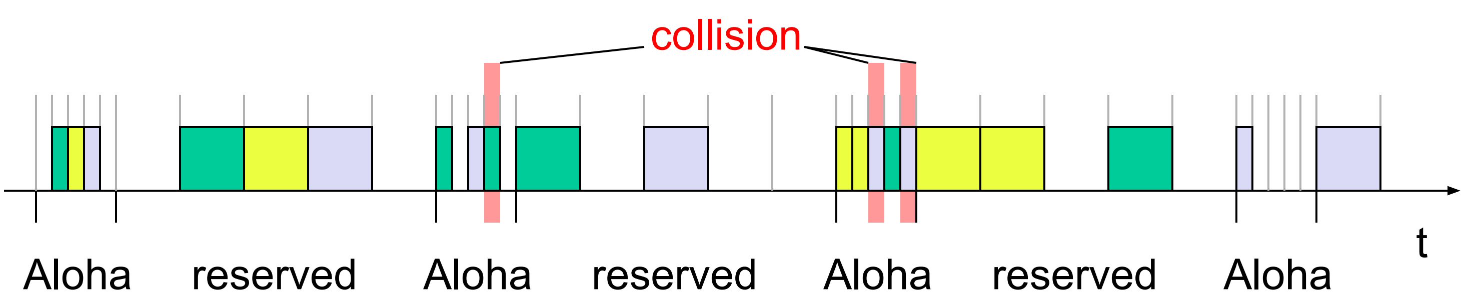
Examples for reservation algorithms:

- *Explicit Reservation according to Roberts (Reservation-ALOHA)*
- *Implicit Reservation (PRMA)*
- *Reservation-TDMA*

Access method DAMA: Explicit Reservation

Explicit Reservation (Reservation Aloha):

- two modes:
  - *ALOHA mode* for reservation:
    competition for small reservation slots, collisions possible
  - *reserved mode* for data transmission within successful reserved slots (no collisions possible)
- it is important for all stations to keep the reservation list consistent at any point in time and, therefore, all stations have to synchronize from time to time



Access method DAMA: PRMA

Implicit reservation (PRMA - Packet Reservation MA):

- a certain number of slots form a frame, frames are repeated
- stations compete for empty slots according to the slotted aloha principle
- once a station reserves a slot successfully, this slot is automatically assigned to this station in all following frames as long as the station has data to send
- competition for this slots starts again as soon as the slot was empty in the last frame

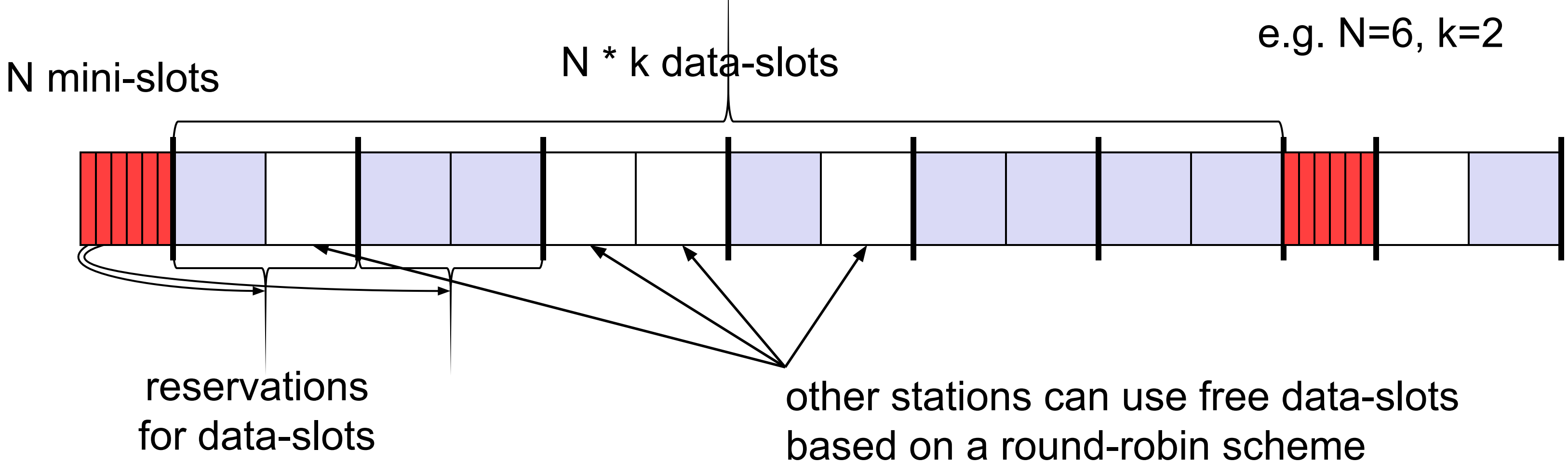
reservation		1	2	3	4	5	6	7	8	time-slot
ACDABA-F	frame$_1$	A	C	D	A	B	A		F	
ACDABA-F	frame$_2$	A	C		A	B	A			
AC-ABAF-	frame$_3$	A				B	A	F		
A---BAFD	frame$_4$	A				B	A	F	D	
ACEEBAFD	frame$_5$	A	C	E	E	B	A	F	D	t

collision at reservation attempts

Access method DAMA: Reservation-TDMA

Reservation Time Division Multiple Access

- every frame consists of N mini-slots and x data-slots
- every station has its own mini-slot and can reserve up to k data-slots using this mini-slot (i.e. x = N * k).
- other stations can send data in unused data-slots according to a round-robin sending scheme (best-effort traffic)



MACA - collision avoidance

MACA (Multiple Access with Collision Avoidance) uses short signaling packets for collision avoidance

- RTS (request to send): a sender request the right to send from a receiver with a short RTS packet before it sends a data packet
- CTS (clear to send): the receiver grants the right to send as soon as it is ready to receive

Signaling packets contain

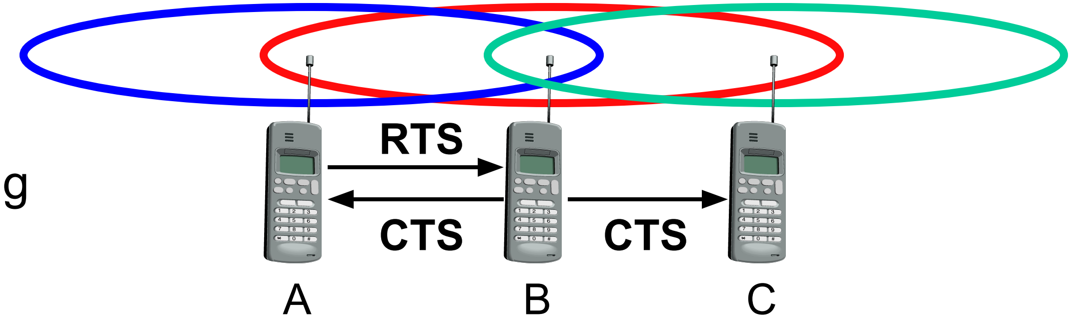
- sender address
- receiver address
- packet size

Variants of this method can be found in IEEE802.11 as DFWMAC (Distributed Foundation Wireless MAC)

MACA examples

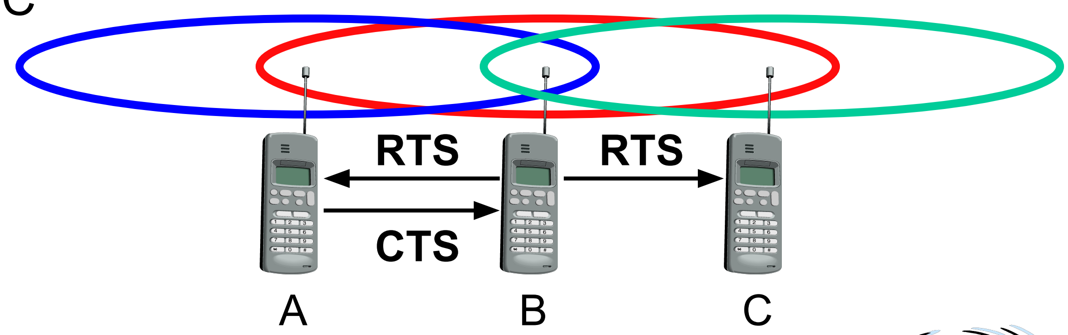
MACA avoids the problem of hidden terminals

- A and C want to send to B
- A sends RTS first
- C waits after receiving CTS from B

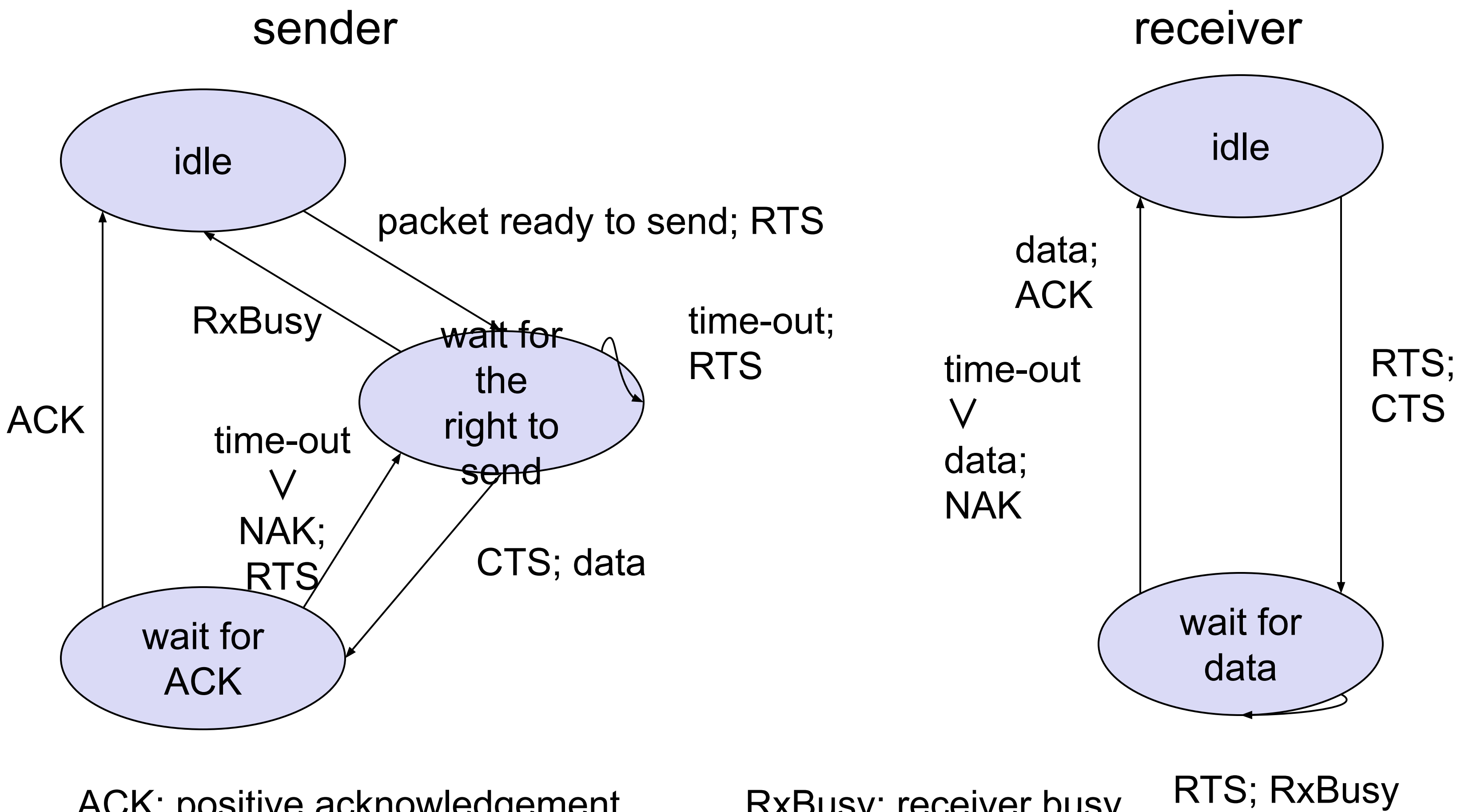


MACA avoids the problem of exposed terminals

- B wants to send to A, C to another terminal
- now C does not have to wait for it cannot receive CTS from A



MACA variant: DFWMAC in IEEE802.11



ACK: positive acknowledgement
NAK: negative acknowledgement

RxBusy: receiver busy

Polling mechanisms

If one terminal can be heard by all others, this "central" terminal (a.k.a. base station) can poll all other terminals according to a certain scheme

- now all schemes known from fixed networks can be used (typical mainframe - terminal scenario)

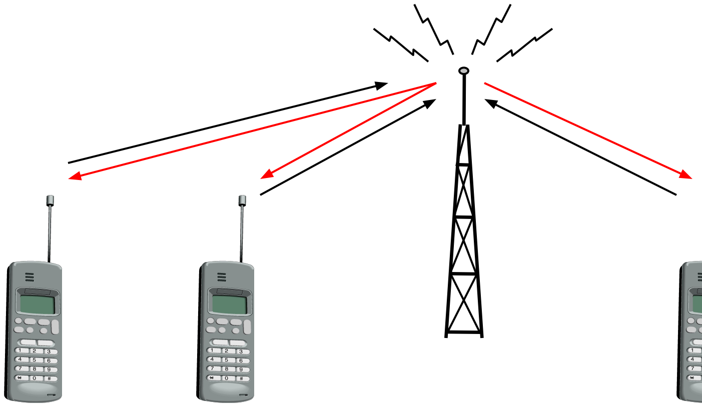
Example: Randomly Addressed Polling

- base station signals readiness to all mobile terminals
- terminals ready to send can now transmit a random number without collision with the help of CDMA or FDMA (the random number can be seen as dynamic address)
- the base station now chooses one address for polling from the list of all random numbers (collision if two terminals choose the same address)
- the base station acknowledges correct packets and continues polling the next terminal
- this cycle starts again after polling all terminals of the list

ISMA (Inhibit Sense Multiple Access)

Current state of the medium is signaled via a “busy tone”

- the base station signals on the downlink (base station to terminals) if the medium is free or not
- terminals must not send if the medium is busy
- terminals can access the medium as soon as the busy tone stops
- the base station signals collisions and successful transmissions via the busy tone and acknowledgements, respectively (media access is not coordinated within this approach)
- mechanism used, e.g., for CDPD (USA, integrated into AMPS)



Access method CDMA

CDMA (Code Division Multiple Access)

- all terminals send on the same frequency probably at the same time and can use the whole bandwidth of the transmission channel
- each sender has a unique random number, the sender XORs the signal with this random number
- the receiver can “tune” into this signal if it knows the pseudo random number, tuning is done via a correlation function

Disadvantages:

- higher complexity of a receiver (receiver cannot just listen into the medium and start receiving if there is a signal)
- all signals should have the same strength at a receiver

Advantages:

- all terminals can use the same frequency, no planning needed
- huge code space (e.g. 2^{32}) compared to frequency space
- interferences (e.g. white noise) is not coded
- forward error correction and encryption can be easily integrated

What is a good code for CDMA?

- Code for a certain user should
  - have a good autocorrelation and
  - be **orthogonal** to other codes
- Two vectors are called orthogonal if their inner product is 0, as is the case for the two vectors (2, 5, 0) and (0, 0, 17): (2, 5, 0)*(0, 0, 17) = 0 + 0 + 0 = 0. But also vectors like (3, –2, 4) and (–2, 3, 3) are orthogonal: (3, –2, 4)*(–2, 3, 3) = –6 – 6 + 12 = 0. By contrast, the vectors (1,2,3) and (4,2, –6) are not orthogonal (the inner product is –10), and (1, 2, 3) and (4, 2, –3) are “almost” orthogonal, with their inner product being –1 (which is “close” to zero).
- The Barker code (+1, –1, +1, +1, –1, +1, +1, +1, –1, –1, –1), for example, has a good autocorrelation, i.e., the inner product with itself is large, the result is 11.

CDMA in theory

Sender A

- sends $A_d = 1$, key $A_k = 010011$ (assign: „0“= -1, „1“= +1)
- sending signal $A_s = A_d * A_k = (-1, +1, -1, -1, +1, +1)$

Sender B

- sends $B_d = 0$, key $B_k = 110101$ (assign: „0“= -1, „1“= +1)
- sending signal $B_s = B_d * B_k = (-1, -1, +1, -1, +1, -1)$

Both signals superimpose in space

- interference neglected (noise etc.)
- $A_s + B_s = (-2, 0, 0, -2, +2, 0)$

Receiver wants to receive signal from sender A

- apply key A_k bitwise (inner product)
  - $A_e = (-2, 0, 0, -2, +2, 0) \cdot A_k = 2 + 0 + 0 + 2 + 2 + 0 = 6$
  - result greater than 0, therefore, original bit was „1“
- receiving B
  - $B_e = (-2, 0, 0, -2, +2, 0) \cdot B_k = -2 + 0 + 0 - 2 - 2 + 0 = -6$, i.e. „0“

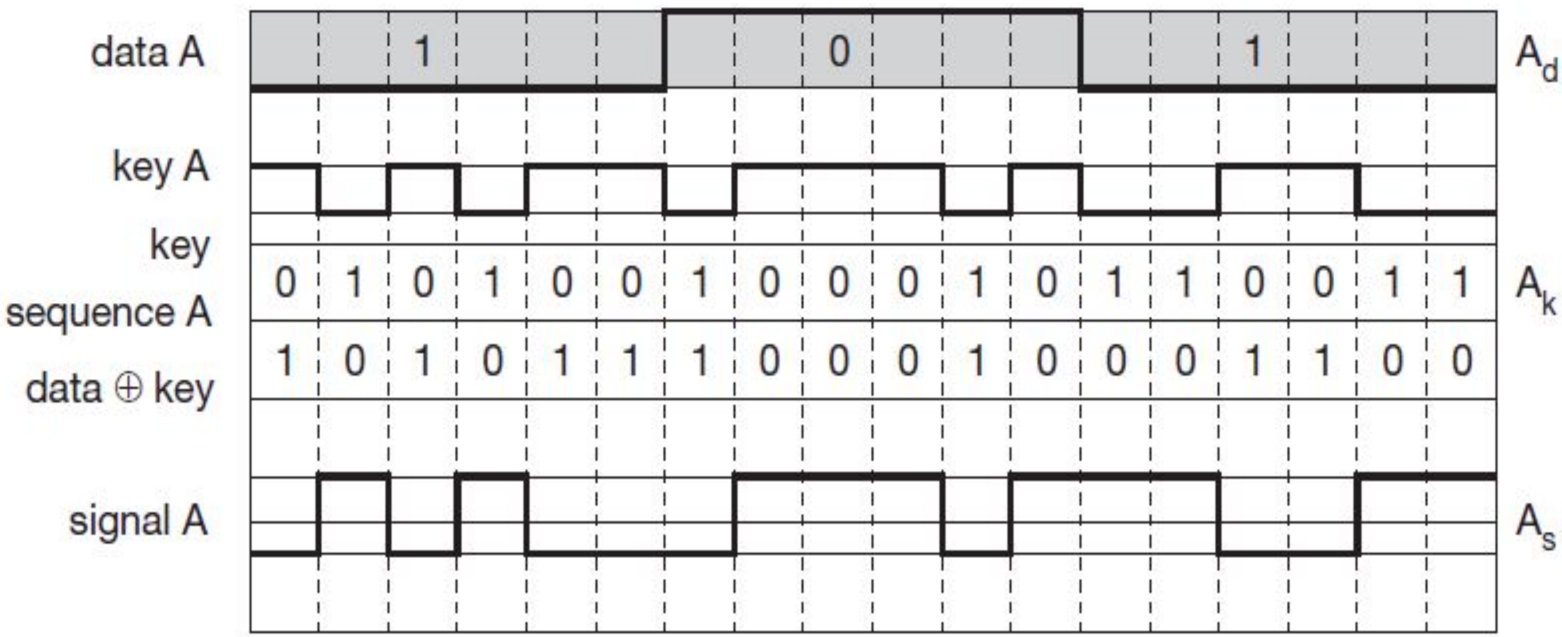
CDMA in theory

- involved several simplifications. The codes were extremely
  - ❑ simple, but at least orthogonal.
  - ❑ More importantly, noise was neglected.
- Noise would add to the transmitted signal C, the results would not be as even with –6 and +6, but would maybe be close to 0, making it harder to decide if this is still a valid 0 or 1.
- Additionally, both spread bits were precisely superimposed and both signals are equally strong when they reach the receiver.

CDMA in theory

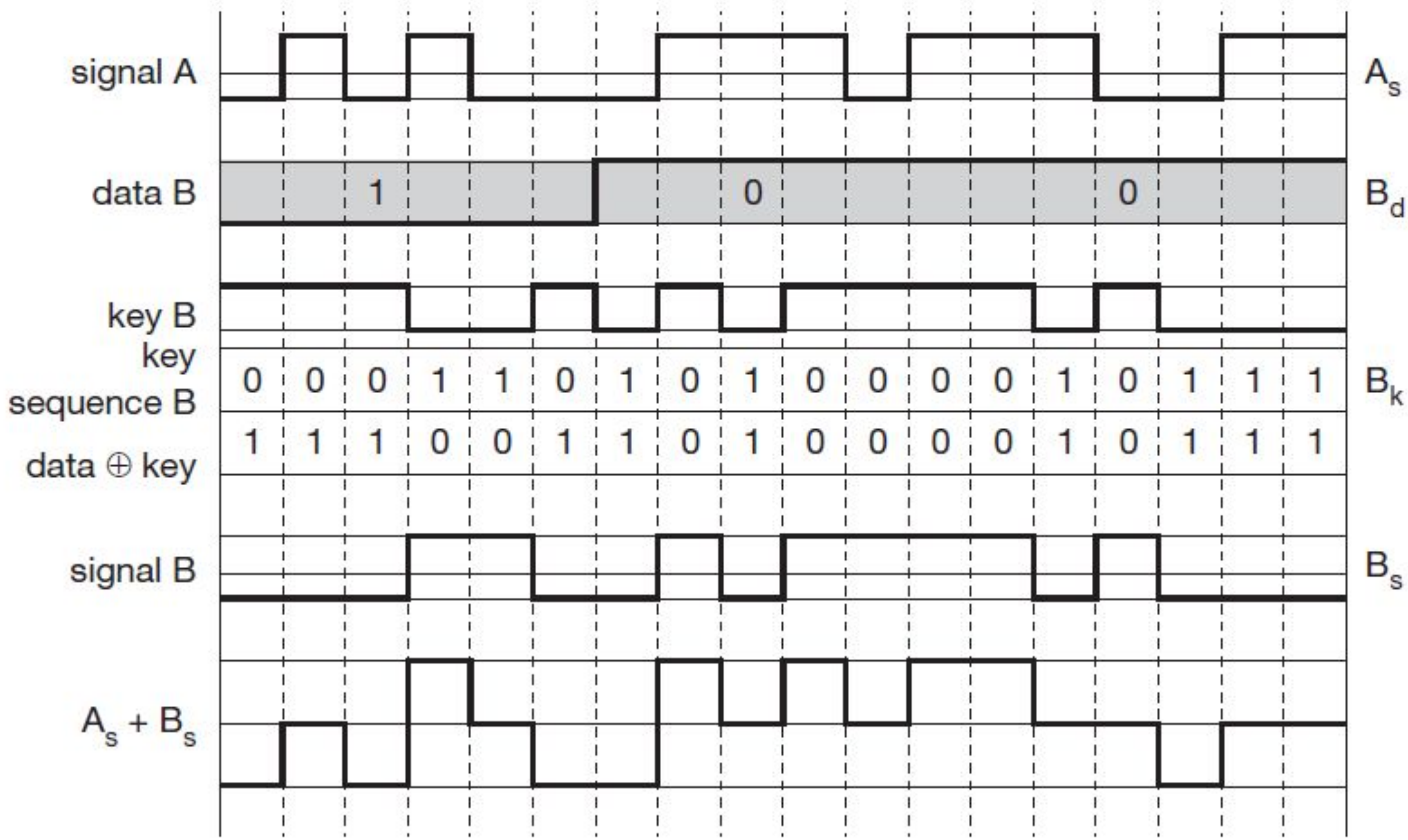
- **What would happen if, for example, B was much stronger?**
  - Assume that B’s strength is five times A’s strength. Then, C' = As + 5*Bs = (–1, +1, –1, –1, +1, +1) + (–5, –5, +5, –5, +5, –5) = (–6, –4, +4, –6, +6, –4).
  - Again, a receiver wants to receive B: C'*Bk = –6 – 4 – 4 – 6 – 6 – 4= –30. It is easy to detect the binary 0 sent by B.
  - Now the receiver wants to receive A: C'*Ak = 6 – 4 – 4 + 6 + 6 – 4 = 6.
  - Clearly, the (absolute) value for the much stronger signal is higher (30 compared to 6).
  - While –30 might still be detected as 0, this is not so easy for the 6 because compared to 30, 6 is quite close to zero and could be interpreted as noise.

Coding and spreading of data from sender A

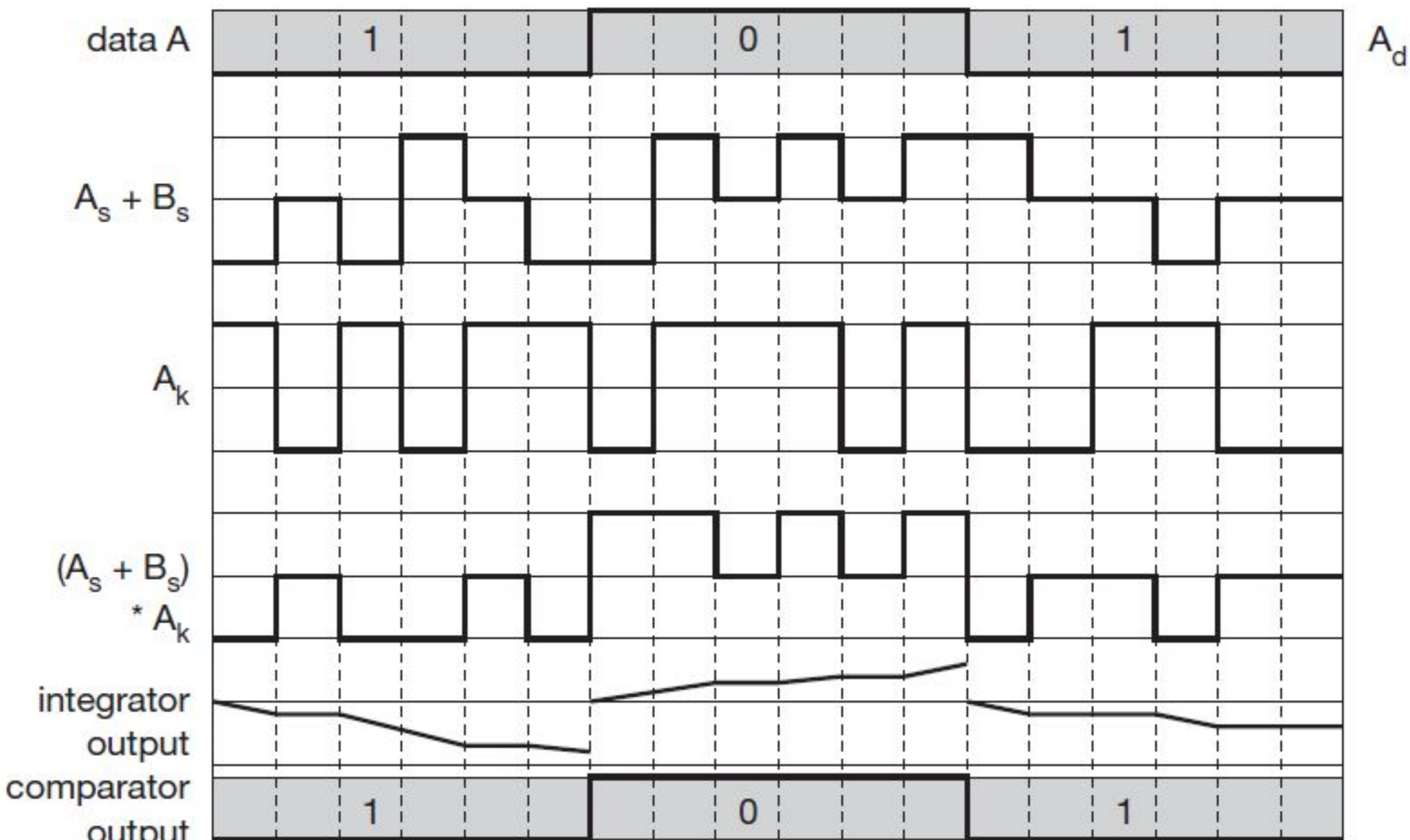


Real systems use much longer keys resulting in a larger distance between single code words in code space.

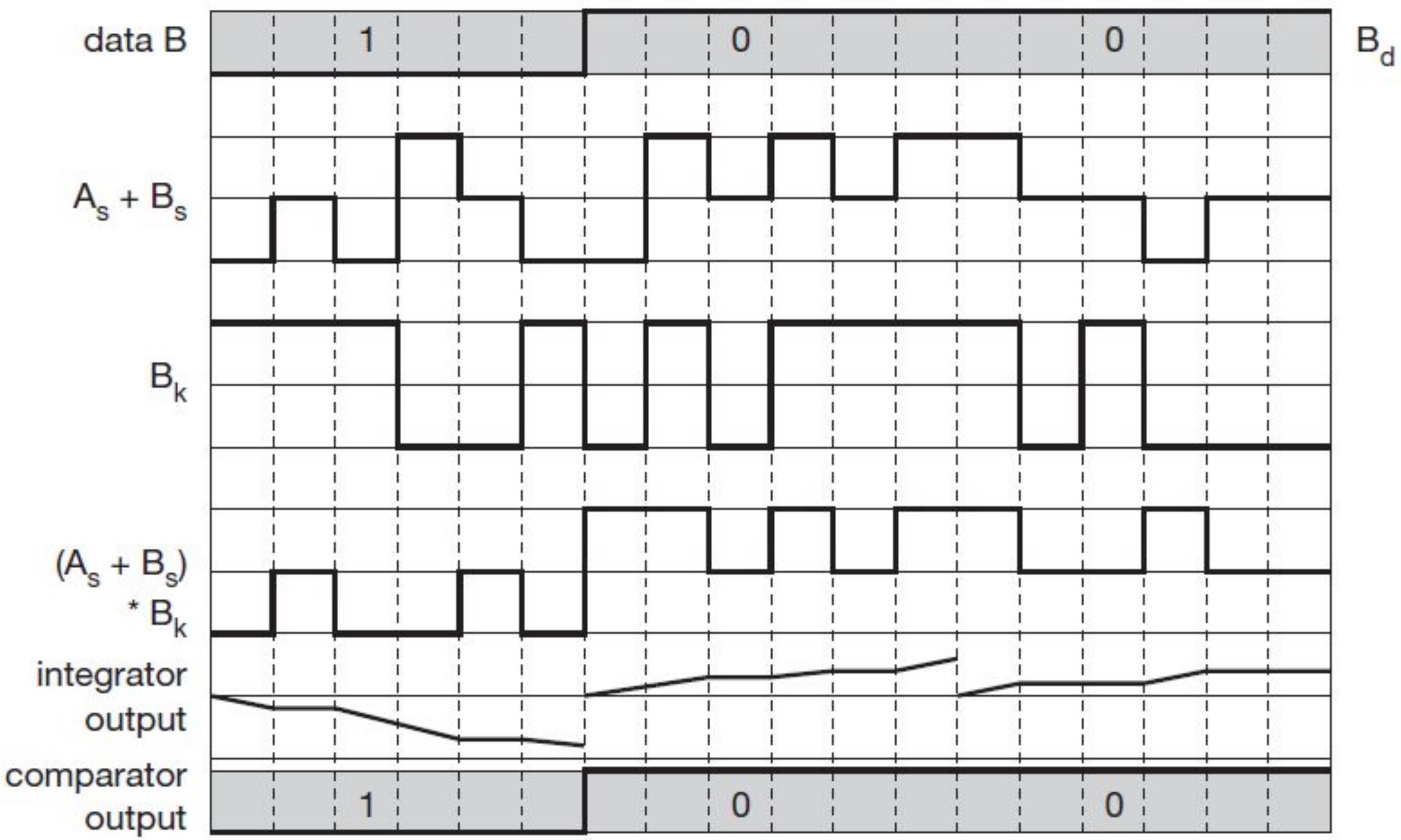
Coding and spreading of data from sender B



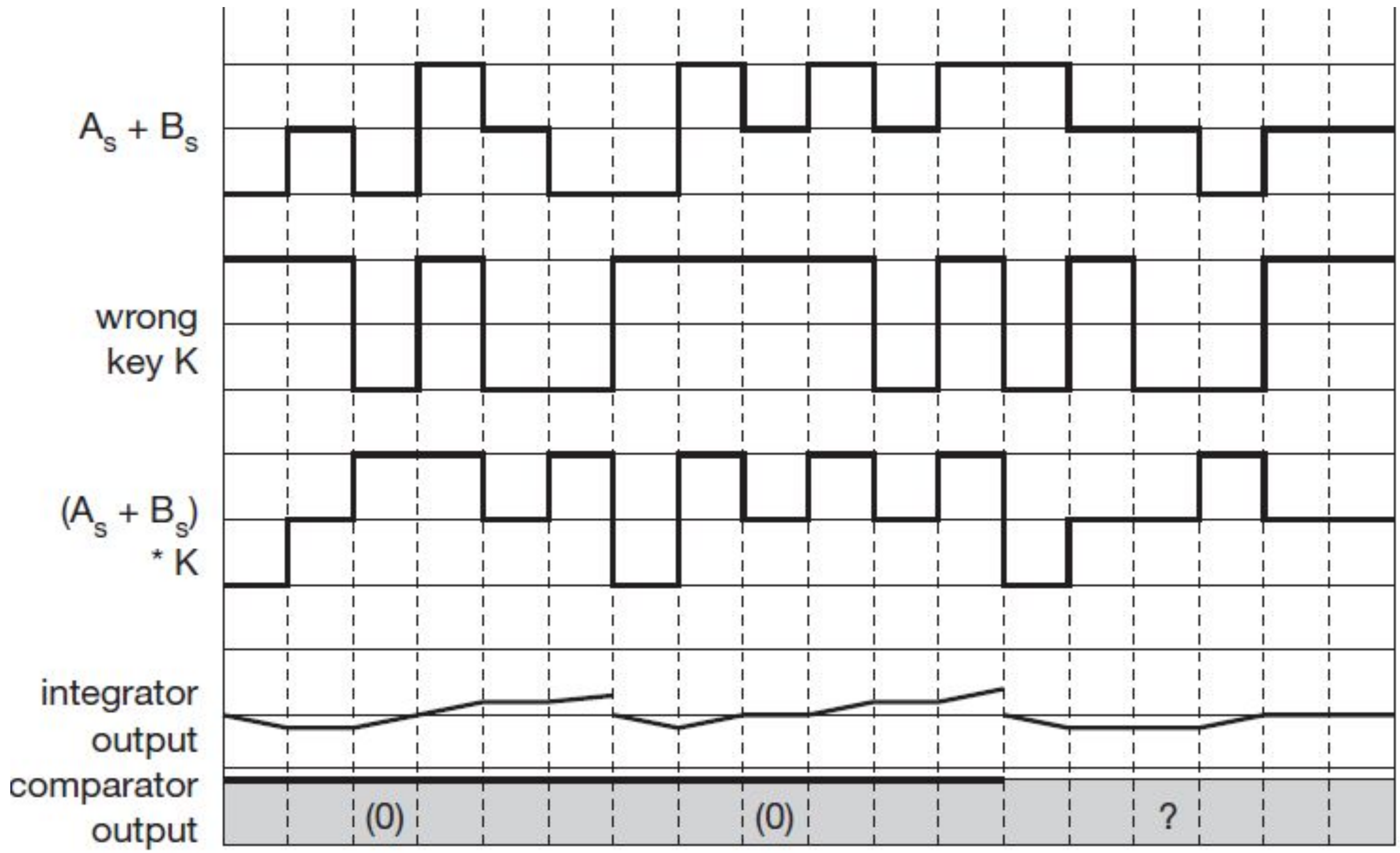
Reconstruction of A's data



Reconstruction of B's data



Receiving a signal with the wrong key



- **How the chipping sequences are synchronized with each sender before sending packets in case of CDMA?**

1. What “synchronization” means in CDMA

- In CDMA, each sender uses a **unique chipping sequence** (spreading code) to spread its data bits into many smaller **chips**.
- Synchronization means:
  - ❑ The **receiver knows exactly which chipping sequence** a sender is using
  - ❑ The receiver is **time-aligned** with that sequence (chip boundaries match)
- Without synchronization, the despreading process won’t work and the data looks like noise.

2. How chipping sequences are synchronized (before data transmission)

- **Step-by-step process**
- **Step 1: Code assignment**
- Each sender is assigned a **unique spreading code**, typically:
  - ❑ Walsh codes (synchronous CDMA, downlink)
  - ❑ PN (pseudo-noise) sequences (asynchronous CDMA, uplink)
- These codes are:
  - ❑ Known to both sender and receiver
  - ❑ Designed to be orthogonal or nearly orthogonal

- In CDMA, **synchronization** means aligning the **chipping sequences in time** so that spreading and despreading work correctly.
- The key difference:
  - **Downlink (base station → users): Synchronous CDMA**
  - **Uplink (users → base station): Asynchronous CDMA**
- This difference exists because of **who controls timing**.

Downlink synchronization (Forward link)

- **Who transmits?**
  - **One base station**, many users
- **Why synchronization is easy**
  - All signals originate from the **same transmitter**
  - Base station can align **all users' spreading codes perfectly in time**
- **How it works**
  - Base station transmits using **orthogonal codes** (e.g., Walsh codes)
  - All users receive signals that are **chip-synchronous**

- Each mobile:
  - ❑ Knows its assigned code
  - ❑ Is already synchronized to the base station’s timing
- **Result**
  - ✔ Perfect orthogonality (ideally)
    - ✔ Minimal multi-user interference

Example (downlink)

- Two users receive:
- User 1 code:
  - $C1=[+1,+1,-1,-1]$
- User 2 code:
  - $C2=[+1,-1,+1,-1]$
- Because the base station aligns them:
  - $C1 \cdot C2=0$
- So:
- User 1 sees **no interference** from User 2
- User 2 sees **no interference** from User 1
  - This is **synchronous CDMA**.

Uplink synchronization (Reverse link)

- **Who transmits?**
  - ❑ **Many users**, one base station
- **Why synchronization is hard**
  - ❑ Users are at **different distances**
  - ❑ Signals arrive with **different delays**
  - ❑ Each user has its **own clock**
- So even if users transmit at the “same time”:
  ➡ the base station receives them **misaligned**

How uplink synchronization works?

- Users transmit using **PN (pseudo-noise) sequences**
- Codes are **not perfectly orthogonal**
- Base station:
  - Separately synchronizes to **each user**
  - Uses **correlators** to lock onto each user’s code and timing
- This is called **asynchronous CDMA**.

Example (uplink)

- Assume:
  - User A signal arrives **on time**
  - User B signal arrives **1 chip late**
- Even if they use orthogonal codes originally, at the base station:
  - $CA(t) \cdot CB(t-\tau) \neq 0$
- Orthogonality is lost
  ➡ Some **multiple access interference (MAI)** exists
- This is why uplink is interference-limited.

Synchronization process comparison

Feature	Downlink	Uplink
Direction	Base → Users	Users → Base
Type	Synchronous CDMA	Asynchronous CDMA
Timing control	Centralized (base station)	Distributed (many users)
Code type	Walsh / orthogonal codes	PN sequences
Orthogonality	Preserved	Lost due to delays
Interference	Very low	Higher (MAI)
Synchronization effort	Easy	Complex

What “asynchronous CDMA” means

- In **asynchronous CDMA**,
  different users’ chipping sequences arrive at the receiver **with different time offsets**, so orthogonality is destroyed.

This is exactly what happens on the **uplink**.

- **Two users transmitting to one base station**
- User A code
  - $CA=[+1, +1, -1, -1]$
- User B code
  - $CB=[+1, -1, +1, -1]$
- These codes are **orthogonal if aligned**:
  - $CA \cdot CB=0$

Assume:

- 1 data bit per code
- Bit $1 \rightarrow +1$
- Bit $0 \rightarrow -1$

- **Transmitted signals (perfectly fine at transmitters)**
- **User A sends bit 1**
  - ❑ $SA=+1\times CA=[+1, +1, -1, -1]$
- **User B sends bit 1**
  - ❑ $SB=+1\times CB=[+1, -1, +1, -1]$

- **Asynchronous arrival at the base station**
- Now the key point:
  - User A arrives **on time**
  - User B arrives **1 chip late**
- So the received version of User B is **shifted**:
  - SB(shifted)=[0, +1, −1, +1]
- (0 means no chip yet)

- **Composite received signal**
- Add the signals chip by chip:

Chip index	User A	User B (shifted)	Sum
1	+1	0	+1
2	+1	+1	+2
3	−1	−1	−2
4	−1	+1	0

$$R = [+1,\ +2,\ -2,\ 0]$$

- **Despreading for User A (this is where async hurts)**
- The base station tries to recover **User A** using CA:
- **Multiply:**
- $R \times CA = [+1, +2, -2, 0] \times [+1, +1, -1, -1] = [+1, +2, +2, 0]$
- **Integrate (sum):**
- $1+2+2+0=5$
- User A’s bit is still detected as **1**, but notice:
- The value should ideally be **4**
- Extra energy came from **User B interference**
  - ❑ This extra term is **Multiple Access Interference (MAI)**.

- **Despreading for User B (even worse)**
- Receiver correlates with **unshifted** CB:
  - R×CB=[+1, +2, −2, 0]×[+1, −1, +1, −1]
    =[+1, −2, −2, 0]
  - Sum: 1−2−2=−3
  - Receiver may detect the **wrong bit**, depending on noise.

- **Why orthogonality is lost**
- Orthogonality requires:
  - $\int CA(t) \cdot CB(t)\, dt = 0$
- But in asynchronous CDMA:
  - $\int CA(t) \cdot CB(t-\tau)\, dt \neq 0$
- Any non-zero delay τ(tau):
  - Breaks orthogonality
  - Creates MAI

Visual intuition (timing)

```
User A: | +1 | +1 | -1 | -1 |
User B:      | +1 | -1 | +1 | -1 |
               ↑ 1-chip delay
```

- The chips **don’t line up**, so cancellation fails.

Key consequences of asynchronous CDMA

Effect	Result
Chip misalignment	Orthogonality lost
Cross-correlation	Non-zero
Interference	MAI
Capacity	Interference-limited
Power control	Critical

- This is **why CDMA uplink needs**:
  - Tight power control
  - Complex receivers (RAKE, multiuser detection)

Why tight power control is needed (near–far problem)

In **asynchronous CDMA uplink**:

- Users arrive at the base station with **different powers**
- A nearby user can **overwhelm** a far user
- Because codes are not perfectly orthogonal → **MAI**

Goal:

Make **all users arrive at the base station with nearly equal received power**

This is what “tight” means.

- **Two-level power control in CDMA**
- CDMA uses **two loops**:
- **Open-loop power control** (coarse)
- **Closed-loop power control** (fine, fast, tight)
- Both are needed.

Open-loop power control (initial estimate)

- **Purpose**
- Compensate for **path loss** before transmission starts.
- **How it works**
- Mobile measures **downlink power** from base station
- Assumes uplink path loss is similar
- Sets initial transmit power:
  - **P**tx=**P**target+Path loss estimate
- **Result**
- Rough power alignment
- Not accurate enough (fading, shadowing differ)
- 📌 This is **not tight** — just a starting point.

Closed-loop power control (this makes it TIGHT)

- **Purpose**
- Continuously fine-tune uplink power **while transmitting**
- **How it works (step-by-step)**
- Base station measures **received SIR** (Signal-to-Interference Ratio)
- Compares it with **target SIR**
- Sends a **1-bit command**:
  - “Power up”
  - “Power down”
- Mobile adjusts transmit power by a **small step**

- Typical CDMA numbers (important)
  - ❑ Update rate: 800–1500 times per second
  - ❑ Step size: ±1 dB
  - ❑ Error margin: < 1 dB
- That’s why it’s called tight.
- Numerical example (near–far fix)
- Without power control
  - ❑ UE1 (near): arrives at −60 dBm
  - ❑ UE2 (far): arrives at −90 dBm
- Difference = 30 dB → UE2 is drowned ❌

- **With tight power control**
- Target received power:

$$P_{target} = -80 \text{ dBm}$$

UE	Initial	Adjustment	Final
Near	−60 dBm	↓ 20 dB	−80 dBm
Far	−90 dBm	↑ 10 dB	−80 dBm

- ✔ Both users now contribute **equal interference**
  ✔ Receiver can despread correctly

- **Why power control must be FAST**
- In asynchronous CDMA:
  - ❑ Channel changes rapidly (fading)
  - ❑ Interference depends on **all users**
- If power control is slow:
  - ❑ One user spikes → raises noise floor
- Everyone’s performance collapses
- 📌 This is why CDMA capacity is **interference-limited**.

Why CDMA capacity collapses without power control

- CDMA is **interference-limited**, not bandwidth-limited. All users transmit **at the same time on the same frequency band**, and they're separated only by their spreading codes. Because of that: **Every user is interference to every other user.**
- **1. The near–far problem**
- Without power control:

A user **close to the base station** arrives with very high power

A user **far away** arrives with much lower power

- The strong (near) signal **overwhelms** the weak (far) signal at the receiver.
- Even though the codes are different, they are **not perfectly orthogonal** in real channels, so:

Strong users raise the interference floor

Weak users' signals fall below detectable levels

- Result: **far users drop out first**

- **2. Interference grows superlinearly**
- In CDMA, capacity depends on the **signal-to-interference ratio (SIR)**:
- [\text{SIR} \approx \frac{P_\text{user}}{\sum P_\text{others}}]
- Without power control:

One loud user increases interference for **everyone**

Other users try to compensate by increasing power

This causes **runaway interference**

- This positive feedback loop leads to **rapid capacity collapse**, not gradual degradation.

- **3. Power control equalizes received power**
- With fast, tight power control:
- Every user arrives at the base station with **approximately the same power**
- No user dominates the interference
- SIR remains predictable
- Capacity scales smoothly with number of users
- Without it:
- A few strong users can destroy capacity for many others
- The system becomes unstable

Bottom line

- **CDMA only works because of power control.**
- Without power control:

Near–far problem dominates

Interference explodes

Users drop out rapidly

System capacity collapses

- That’s why CDMA systems (IS-95, cdma2000, WCDMA) require **very fast closed-loop power control**—often hundreds to thousands of adjustments per second.

- **Explain why TDMA/FDMA don’t suffer this collapse as severely**

What “collapse” means in CDMA

- In **CDMA**, all users transmit **at the same time and in the same band**, separated only by spreading codes. Capacity is **interference-limited**, not resource-limited.
- As you add users:
- Each new user raises the **noise floor** for everyone else.
- To maintain SIR, all users increase power.
- That extra power raises interference again → everyone boosts power again.
- This **positive feedback loop** can run away. Past a certain load, the system can’t converge to stable power levels, SIR drops for everyone, and **all calls degrade or fail at once** → *capacity collapse*.
- This is the classic *“soft capacity cliff.”*

Why TDMA / FDMA behave differently

- **1. Orthogonal resource separation**
- **FDMA:** users get separate frequency bands
- **TDMA:** users get separate time slots
- Ideally, users are **orthogonal**, so:
- One user’s signal does *not* raise the interference seen by others
- Adding a user does not degrade existing users (until resources run out)
- Interference mainly comes from:
- Adjacent channel leakage
- Imperfect filters
- External interferers
- Not from *other active users in the same cell*.

- **2. Hard capacity limit**
- TDMA/FDMA have a **fixed number of channels or slots**.
- Once they’re full:
- New users are **blocked**
- Existing users keep their QoS
- No feedback loop that drags everyone down
- So instead of collapse:
- You get **graceful blocking**, not system-wide degradation.

- **3. No near–far amplification loop**
- CDMA is extremely sensitive to the **near–far problem**:
- Strong users can drown out weak ones
- Power control errors affect *everyone*
- TDMA/FDMA:
- Power control errors are mostly **local**
- A loud user ruins *their own* channel, not the entire cell
- Failures stay **contained**.

- **4. Interference doesn’t scale with load**
- In TDMA/FDMA:
- Interference is roughly constant as load increases
- Performance stays stable until resources are exhausted
- In CDMA:
- Interference scales roughly **linearly with the number of users**
- Performance degrades *continuously* with load

- **Visual mental model**
- **TDMA / FDMA**
- “Hotel rooms”
  When all rooms are booked, guests are turned away.
  People already inside sleep just fine.
- **CDMA**
- “Everyone shouting in the same room”
  Each new person forces everyone to shout louder
  Eventually, nobody can be understood.

Feature	TDMA / FDMA	CDMA
Capacity type	Hard	Soft
User separation	Time / Frequency	Codes
Interference growth with load	Low	High
Failure mode	Call blocking	System-wide collapse
Sensitivity to power control	Low	Very high

Why CDMA Uplink Needs Complex Receivers

- In the **uplink**:
- Users are at **different distances** from the base station
- Signals arrive with **different powers**
- Channels are **multipath fading**
- Signals are **not perfectly orthogonal** (unlike downlink)
- So the base station receives a **superposition** of:
- Multiple users
- Multiple delayed versions (multipath) of each user
- Noise
- This creates:
- **Near–far problem**
- **Multiple Access Interference (MAI)**
- **Inter-Symbol Interference (ISI)**

- To deal with this, CDMA relies on:

1. **RAKE receivers** → handle multipath
2. **Multiuser detection (MUD)** → handle interference from other users

1. RAKE Receiver in CDMA

- **1.1 Problem: Multipath in CDMA**
- In wireless channels, a transmitted signal arrives as multiple delayed copies:

$$r(t) = \sum_{l=1}^{L} a_l s(t - \tau_l) + n(t)$$

- In narrowband systems → multipath causes **fading**
- In CDMA → multipath can be **exploited**, because:
- Chip duration is very small
- Different paths are often **resolvable in time**

- **1.2 What Is a RAKE Receiver?**
- A **RAKE receiver**:
- Uses **multiple correlators**, called **fingers**
- Each finger is synchronized to a different multipath component
- Combines energy from all significant paths
- Think of it as “raking” signal energy from different delays

- **1.3 RAKE Receiver Structure**
- Each RAKE finger does:

1. **Delay alignment**
   Aligns to a specific path delay τl
2. **Despreading**
   Correlates the received signal with the user’s spreading code
3. **Channel weighting**
   Multiplies by estimated channel gain al∗a_l^*al∗
4. **Combining**
   Outputs from all fingers are summed

- Mathematically:

$$\hat{s}(t) = \sum_{l=1}^{L} a_l^* \int r(t)c(t-\tau_l)dt$$

- Common combining methods:
- **Maximal Ratio Combining (MRC)** (most common)
- Equal Gain Combining (EGC)
- Selection Combining

- **1.4 Why RAKE Is Needed in Uplink**
- Multipath delays differ per user
- Base station must resolve and track each user’s channel
- More users → more multipath components → more fingers
- ➡️ **Receiver complexity grows rapidly**

- **2. Multiuser Detection (MUD) in CDMA**
- **2.1 Problem: Multiple Access Interference (MAI)**
- In uplink CDMA:
- Spreading codes are **not perfectly orthogonal**
- Signals arrive with different powers
- Each user appears as **noise** to others
- Received signal:

$$r(t) = \sum_{k=1}^{K} \sqrt{P_k} b_k c_k(t) + n(t)$$

- A simple receiver treats MAI as noise → **suboptimal**

- **2.2 What Is Multiuser Detection?**
- **Multiuser detection** jointly detects multiple users instead of detecting one user at a time.
- Goal:
- Separate users **by signal processing**, not just by codes
- **2.3 Types of Multiuser Detectors**
- **(a) Optimal Multiuser Detector (MLD)**
- Uses **maximum likelihood detection**
- Considers all possible bit combinations
- Complexity:

$$O(2^K)$$

- ❌ Impractical for real systems

- **(b) Linear Multiuser Detectors**

1. **Decorrelator**
   1. Inverts code correlation matrix
   2. Removes MAI completely (no noise consideration)
   3. Amplifies noise
2. **MMSE Detector**
   1. Balances noise and MAI suppression
   2. Most practical linear MUD

- MMSE filter:

$$\mathbf{w}_k = (\mathbf{R} + \sigma^2 \mathbf{I})^{-1} \mathbf{c}_k$$

- **(c) Interference Cancellation Techniques**
- These are very common in CDMA uplink:
- **i) Successive Interference Cancellation (SIC)**

1. Detect strongest user
2. Reconstruct its signal
3. Subtract it from received signal
4. Repeat for next strongest user

- ✔ Effective with good power control
  ❌ Error propagation risk

- **ii) Parallel Interference Cancellation (PIC)**
- Detect all users simultaneously
- Subtract estimated interference in parallel
- Iterate multiple times
- ✔ Lower delay
  ❌ More computation

- **2.4 Why MUD Is Especially Important in Uplink**
- Power imbalance between users
- Near–far problem
- High system load
- Without MUD:
- Capacity collapses quickly
- Strong users dominate weak users

- **3. Combined Complexity in Uplink CDMA**
- In practice, the base station uses **both**:
  - ❑ Multipath → RAKE receiver
  - ❑ Multiuser interference → MUD
- So for **each user**, the receiver must:
- Track multiple paths
- Estimate channel gains
- Despread signals
- Cancel interference from other users
- ➡️ That's why **uplink CDMA receivers are much more complex than downlink receivers**

- **4. Key Takeaway**
- **CDMA uplink needs complex receivers because:**
- Users share the same spectrum
- Signals arrive with different powers and delays
- Interference is the main capacity limiter
- **RAKE receivers** exploit multipath diversity
  Multiuser detectors suppress or cancel MAI
- Together, they make CDMA uplink feasible—but at the cost of **high receiver complexity**.

Why LTE abandoned CDMA in uplink?

- LTE dropped CDMA in the uplink for a very practical reason: **CDMA's uplink receiver complexity and interference behavior do not scale well when data rates and user counts increase**. LTE's design goals (high data rate, low latency, low UE power consumption, and simple receivers) pushed 3GPP toward **SC-FDMA instead of CDMA**.

1. CDMA Uplink: Fundamental Limitations

- **1.1 Interference-Limited Nature**
- In CDMA uplink:
- All users transmit over the **same time–frequency resources**
- Capacity is **interference-limited**, not bandwidth-limited
- Adding users increases **Multiple Access Interference (MAI)** for everyone
- As system load increases:
  - SINR↓⇒Throughput↓
- This makes **guaranteed high data rates** difficult.

- **1.2 Near–Far Problem → Tight Power Control**
- Users are at different distances from the base station.
- Strong nearby UEs can drown out weak far UEs
- Requires **very fast, very accurate power control** (≈1 dB accuracy)
- Problems:
- Hard to maintain at high mobility
- Control signaling overhead
- Instability under heavy load
- LTE wanted to **relax power-control dependence**, not tighten it.

- **1.3 Receiver Complexity Explosion**
- To make CDMA uplink work well, the eNodeB must use:
- RAKE receivers (multipath resolution)
- Multiuser Detection (MMSE, SIC, PIC)
- Interference cancellation
- Accurate channel estimation per user
- Complexity grows roughly with:
  - Users×Multipath components
- This was acceptable for voice and low-rate data (3G), but **not for broadband packet data**.

2. High Data Rates Don’t Fit CDMA Well

- **2.1 Spreading Gain vs Data Rate Tradeoff**
- CDMA relies on:
  - Spreading Gain=W/R
- To increase data rate R:
- Spreading factor must decrease
- Orthogonality degrades
- Interference increases
- So:
- High data rate users **hurt everyone else**
- Scheduler has little flexibility

- **2.2 Inefficient Resource Allocation**
- CDMA:
- Everyone always transmits
- Hard to assign “just a chunk” of bandwidth to one user
- LTE wanted:
- Fast scheduling
- Frequency-selective allocation
- QoS differentiation
- CDMA fundamentally fights this.

- **3. LTE Design Goals (Why CDMA Didn’t Fit)**
- LTE targeted:
- Peak uplink data rates > 50 Mbps
- Low latency (~1 ms TTI)
- Simple UE transmitter
- Scalable bandwidth (1.4–20 MHz)
- CDMA uplink conflicts with **every one** of these goals.

- **4. Why SC-FDMA Was Chosen Instead**
- LTE uplink uses **SC-FDMA (DFT-spread OFDMA)**.
- **4.1 Lower PAPR (Big Reason)**
- Compared to OFDMA and CDMA:

Scheme	PAPR	UE Power Efficiency
CDMA	High	Poor
OFDMA	Very High	Very Poor
SC-FDMA	Low	Good

- Lower PAPR →
- Cheaper power amplifiers
- Longer battery life
- Less uplink distortion
- This alone was a **killer argument** against CDMA.

- **4.2 Orthogonal Uplink → No MAI**
- In LTE uplink:
- Users are assigned **orthogonal subcarriers**
- No intra-cell MAI (ideally)
- So:
  - ❑ Interference≈Inter-cell only
- This:
- Eliminates near–far problem
- Greatly simplifies receiver
- Reduces need for tight power control

- **4.3 Simple Frequency-Domain Receiver**
- LTE uplink receiver:
- FFT
- One-tap frequency-domain equalization
- Per-user processing
- Compared to:
- RAKE fingers
- Multiuser detection
- Interference cancellation
- ➡️ Massive complexity reduction at the base station.

5. Multipath Handling: RAKE vs OFDM

CDMA	LTE Uplink
Time-domain RAKE	Frequency-domain equalization
Many correlators	Simple FFT
Path tracking	Cyclic prefix removes ISI

- OFDM/SC-FDMA turns multipath from a **problem** into a **feature**.

6. Spectral Efficiency and Scaling

- LTE uplink:
- Scales linearly with bandwidth
- Supports MIMO naturally
- Works well for bursty packet data
- CDMA:
- Interference-limited
- Hard to scale beyond certain load
- Not MIMO-friendly in uplink

- **7. Final Summary (Exam-Ready)**
- **LTE abandoned CDMA in the uplink because CDMA:**

1. Is interference-limited and suffers from MAI
2. Requires tight power control (near–far problem)
3. Needs complex receivers (RAKE + multiuser detection)
4. Scales poorly for high data rates
5. Is inefficient for packet-based scheduling
6. Has high PAPR and poor UE power efficiency

- **SC-FDMA was chosen because it provides:**
- Orthogonal uplink access
- Low PAPR
- Simple receivers
- Better scalability and battery life

Explain why LTE downlink still resembles CDMA ideas

- **1. Key Reason: Downlink Has a Single Transmitter**
- **CDMA Downlink (3G)**
- One transmitter: **base station**
- Multiple users
- Signals can be made **orthogonal at transmission**
- **LTE Downlink**
- Same situation:
  - One eNodeB
  - Many users
- Orthogonality is easy to control
- 👉 This is **the exact condition where CDMA works best**
- The uplink failed because users transmit independently.
  The downlink keeps the CDMA advantage: **centralized control.**

- **2. Orthogonality: Codes vs Subcarriers**
- **CDMA Downlink**
- Users separated by **orthogonal spreading codes** (Walsh codes)
- Orthogonality holds in ideal channels
- **LTE Downlink (OFDMA)**
- Users separated by **orthogonal subcarriers**
- Orthogonality maintained via:
  - FFT
  - Cyclic prefix
- **Same idea, different domain**:

Concept	CDMA	LTE DL
User separation	Codes	Subcarriers
Orthogonality	Walsh codes	FFT tones
Loss of orthogonality	Multipath	CP handles it

- 👉 LTE just moved orthogonality from **code domain → frequency domain**.

- **3. Spreading Still Exists in LTE Downlink**
- LTE downlink still uses **spreading-like mechanisms**:
- **3.1 Resource Element Mapping**
- Each symbol is spread across:
  - Time
  - Frequency
- Increases robustness to fading
- **3.2 Reference Signals (Pilots)**
- Cell-specific reference signals resemble:
  - CDMA pilot channels
- Used for:
  - Channel estimation
  - Coherent detection

- **3.3 Control Channels**
- PDCCH uses:
  - Distributed mapping
  - Interleaving
- Very similar to CDMA’s **interference averaging**

- **4. Frequency Diversity = CDMA’s Processing Gain**
- **CDMA**
- Processing gain averages interference across wide bandwidth
- **LTE Downlink**
- Wideband OFDM + scheduling across subcarriers
- Exploits **frequency diversity**
- Both:
- Trade bandwidth for reliability
- Average out deep fades
- So LTE preserves CDMA’s core principle:
- **Spread energy → gain robustness**

- **5. Interference Treated as Noise (Like CDMA DL)**
- In LTE downlink:
- No multiuser detection
- Other-user signals are **not explicitly cancelled**
- Interference is treated as noise
- Exactly like CDMA downlink receivers:
- Simple matched filters
- No SIC or MUD at UE
- This is intentional:
- Keeps UE receiver **simple**
- Pushes complexity to the network

- **6. Soft Capacity Philosophy Lives On**
- **CDMA Downlink**
- Capacity degrades gracefully with load
- Interference increases smoothly
- **LTE Downlink**
- Adaptive:
  - MCS
  - Scheduling
  - Power allocation
- SINR-based adaptation
- Different tools, same philosophy:
- **Graceful degradation instead of hard blocking**

- **7. MIMO + Precoding = “Spatial CDMA”**
- LTE downlink uses:
- Beamforming
- Precoding
- MU-MIMO
- This is conceptually similar to:
- CDMA’s user separation in another domain

Domain	Separation
CDMA	Code
LTE DL	Frequency + Space

- So LTE extended CDMA thinking into **spatial domain multiplexing**.

- **8. Why LTE Didn’t Use CDMA Directly in Downlink**
- Even though CDMA ideas remain, LTE avoided actual CDMA because:
- OFDM handles multipath much better
- Easier equalization
- Better spectral efficiency
- MIMO-friendly
- So LTE kept:
- **CDMA philosophy**
  But replaced:
- **CDMA implementation**

- **Step 2: Code generation using a known seed**
- Chipping sequences are generated using:
  - Linear feedback shift registers (LFSRs)
  - A known initial state (seed)
- Because the **seed and generator are known**, both sides can generate the **same sequence independently**.

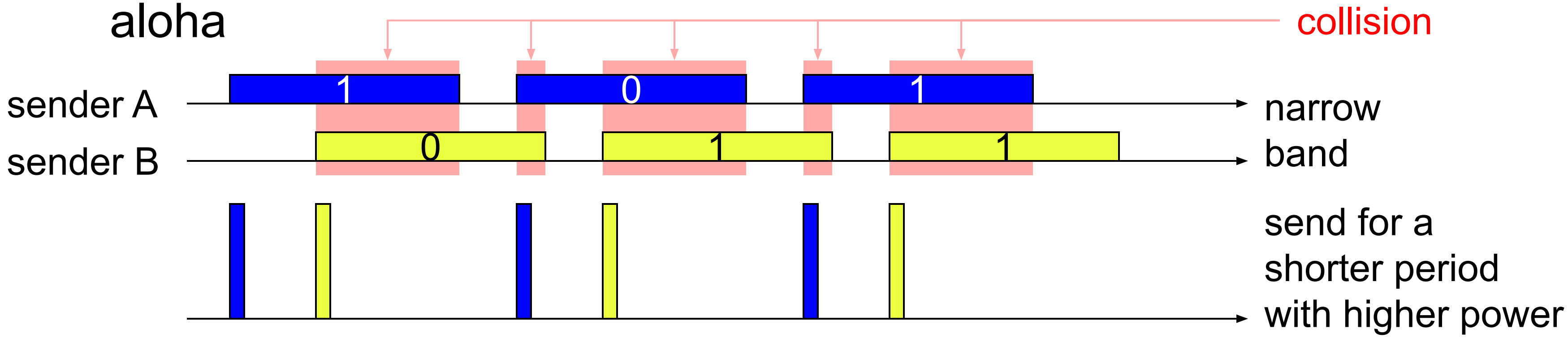
- **Step 3: Timing synchronization**
- Before sending packets, the sender transmits:
  - A **pilot signal** or **preamble**
  - The receiver uses this to:
    - Detect the signal
    - Align chip timing
    - Lock onto the phase of the spreading code
- This is called **code acquisition**.

- **Step 4: Tracking**
- Once acquired, the receiver continuously:
  - ❑ Adjusts timing slightly (using delay-lock loops)
  - ❑ Keeps the code aligned despite drift

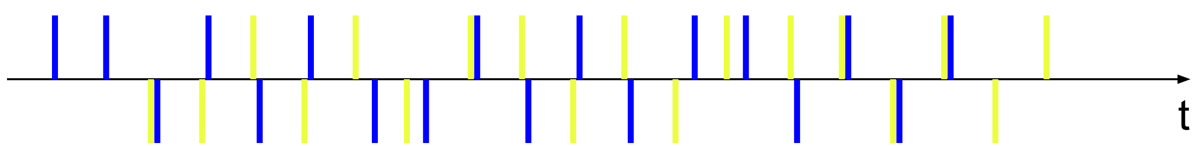
SAMA - Spread Aloha Multiple Access

Aloha has only a very low efficiency, CDMA needs complex receivers to be able to receive different senders with individual codes at the same time

Idea: use spread spectrum with only one single code (chipping sequence) for spreading for all senders accessing according to aloha



spread the signal e.g. using the chipping sequence 110101 („*CDMA without CD*“)



Problem: find a chipping sequence with good characteristics

Comparison SDMA/TDMA/FDMA/CDMA

Approach	SDMA	TDMA	FDMA	CDMA
Idea	segment space into cells/sectors	segment sending time into disjoint time-slots, demand driven or fixed patterns	segment the frequency band into disjoint sub-bands	spread the spectrum using orthogonal codes
Terminals	only one terminal can be active in one cell/one sector	all terminals are active for short periods of time on the same frequency	every terminal has its own frequency, uninterrupted	all terminals can be active at the same place at the same moment, uninterrupted
Signal separation	cell structure, directed antennas	synchronization in the time domain	filtering in the frequency domain	code plus special receivers
Advantages	very simple, increases capacity per km²	established, fully digital, flexible	simple, established, robust	flexible, less frequency planning needed, soft handover
Dis-advantages	inflexible, antennas typically fixed	guard space needed (multipath propagation), synchronization difficult	inflexible, frequencies are a scarce resource	complex receivers, needs more complicated power control for senders
Comment	only in combination with TDMA, FDMA or CDMA useful	standard in fixed networks, together with FDMA/SDMA used in many mobile networks	typically combined with TDMA (frequency hopping patterns) and SDMA (frequency reuse)	still faces some problems, higher complexity, lowered expectations; will be integrated with TDMA/FDMA